

What is organic chemistry?

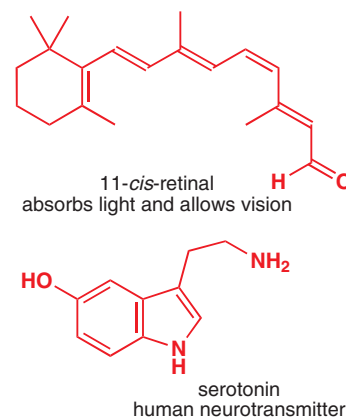
Organic chemistry and you

You are already a highly skilled organic chemist. As you read these words, your eyes are using an organic compound (retinal) to convert visible light into nerve impulses. When you picked up this book, your muscles were doing chemical reactions on sugars to give you the energy you needed. As you understand, gaps between your brain cells are being bridged by simple organic molecules (neurotransmitter amines) so that nerve impulses can be passed around your brain. And you did all that without consciously thinking about it. You do not yet understand these processes in your mind as well as you can carry them out in your brain and body. You are not alone there. No organic chemist, however brilliant, understands the detailed chemical working of the human mind or body very well.

We, the authors, include ourselves in this generalization, but we are going to show you in this book what enormous strides have been taken in the understanding of organic chemistry since the science came into being in the early years of the nineteenth century. Organic chemistry began as a tentative attempt to understand the chemistry of life. It has grown into the confident basis of worldwide activities that feed, clothe, and cure millions of people without their even being aware of the role of chemistry in their lives. Chemists cooperate with physicists and mathematicians to understand how molecules behave and with biologists to understand how interactions between molecules underlie all of life. The enlightenment brought by chemistry in the twentieth century amounted to a revolution in our understanding of the molecular world, but in these first decades of the twenty-first century the revolution is still far from complete. We aim not to give you the measurements of the skeleton of a dead science but to equip you to understand the conflicting demands of an adolescent one.

Like all sciences, chemistry has a unique place in our pattern of understanding of the universe. It is the science of molecules. But organic chemistry is something more. It literally creates itself as it grows. Of course we need to study the molecules of nature both because they are interesting in their own right and because their functions are important to our lives. Organic chemistry has always been able to illuminate the mechanisms of life by making new molecules that give information not available from the molecules actually present in living things.

This creation of new molecules has given us new materials such as plastics to make things with, new dyes to colour our clothes, new perfumes to wear, new drugs to cure diseases. Some people think some of these activities are unnatural and their products dangerous or unwholesome. But these new molecules are built by humans from other molecules found naturally on earth using the skills inherent in our natural brains. Birds build nests; people build houses. Which is unnatural? To the organic chemist this is a meaningless distinction. There are toxic compounds and nutritious ones, stable compounds and reactive ones—but there is only one type of chemistry: it goes on both inside our brains and bodies, and also in our flasks and reactors, born from the ideas in our minds and the skill in our hands. We are not going to set ourselves up as moral judges in any way. We believe it is right to try and understand the world



■ We are going to illustrate this chapter with the structures of the organic compounds we talk about. If you do not understand the diagrams, just read the text. Explanation of the rest is on its way.

about us as best we can and to use that understanding creatively. This is what we want to share with you.

■ At the other end of this book (Chapter 42) you will read about the extraordinary chemistry that allows life to exist—facts that are known only from cooperation between chemists and biologists.

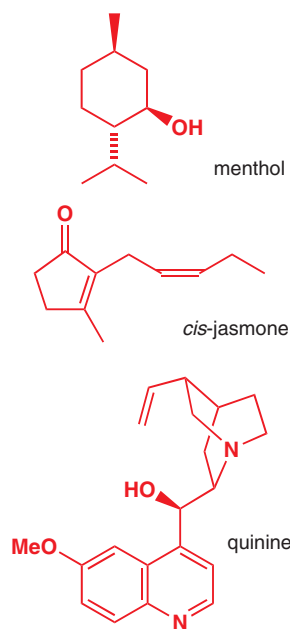
Organic compounds

Organic chemistry started as the chemistry of life, when that was thought to be different from the chemistry in the laboratory. Then it became the chemistry of carbon compounds, especially those found in coal. But now it is both. It is the chemistry of the compounds formed by carbon and other elements such as are found in living things, in the products of living things, and wherever else carbon is found.

The most abundant organic compounds are those present in living things and those formed over millions of years from dead things. In earlier times, the organic compounds known from nature were those in the ‘essential oils’ that could be distilled from plants and the alkaloids that could be extracted from crushed plants with acid. Menthol is a famous example of a flavouring compound from the essential oil of spearmint and *cis*-jasmone an example of a perfume distilled from jasmine flowers.

Natural products have long been used to cure diseases, and in the sixteenth century one became famous—quinine was extracted from the bark of the South American cinchona tree and used to treat fevers, especially malaria. The Jesuits who did this work (the remedy was known as ‘Jesuit’s bark’) did not of course know what the structure of quinine was, but now we do. More than that, the molecular structure of quinine has inspired the design of modern drug molecules which treat malaria much more effectively than quinine itself.

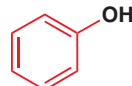
The main reservoir of chemicals available to the nineteenth century chemists was coal. Distillation of coal to give gas for lighting and heating (mainly hydrogen and carbon monoxide) also gave a brown tar rich in aromatic compounds such as benzene, pyridine, phenol, aniline, and thiophene.



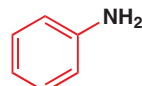
benzene



pyridine



phenol



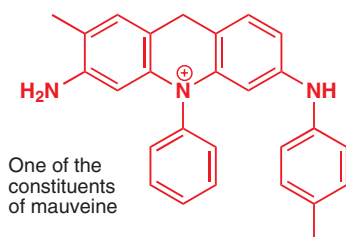
aniline



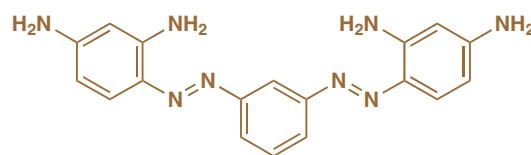
thiophene

Perkin was studying in London with the great German chemist, Hofmann. Perkin’s attempt to make quinine this way was a remarkable practical challenge given that its structure was still unknown.

Phenol was used in the nineteenth century by Lister as an antiseptic in surgery, and aniline became the basis for the dyestuffs industry. It was this that really started the search for new organic compounds made by chemists rather than by nature. In 1856, while trying to make quinine from aniline, an 18-year old British chemist, William Perkin, managed to produce a mauve residue, mauveine, which revolutionized the dyeing of cloth and gave birth to the synthetic dyestuffs industry. A related dyestuff of this kind—still available—is Bismarck Brown: much of the early work on dyes was done in Germany.



One of the constituents of mauveine



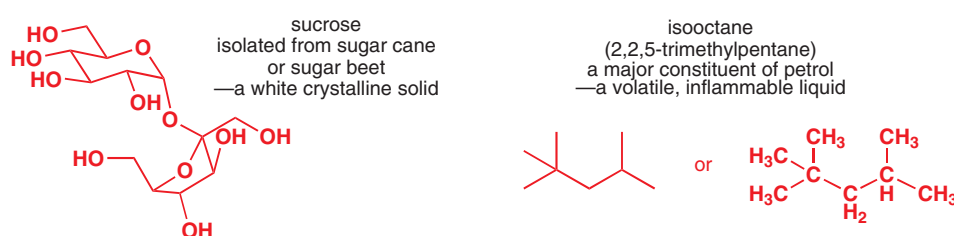
Bismarck Brown Y

In the twentieth century oil overtook coal as the main source of bulk organic compounds so that simple hydrocarbons like methane (CH_4 , ‘natural gas’), propane, and butane ($\text{CH}_3\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$, ‘calor gas’ or LPG) became available for fuel. At the same time chemists began the search for new molecules from new sources such as fungi, corals, and bacteria, and two organic chemical industries developed in parallel—‘bulk’ and

'fine' chemicals. Bulk chemicals like paints and plastics are usually based on simple molecules produced in multitonne quantities while fine chemicals such as drugs, perfumes, and flavouring materials are produced in smaller quantities but much more profitably.

At the time of writing there were over 16 million organic compounds known. How many more might there be? Even counting only moderately sized molecules, containing fewer than about 30 carbon atoms (about the size of the mauveine structure above), it has been calculated that something in the region of 1,000 (10⁶³) stable compounds are possible. There aren't enough carbon atoms in the universe to make them all.

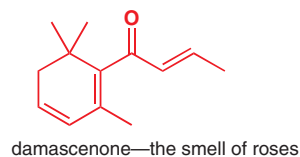
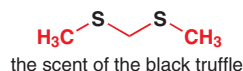
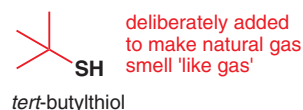
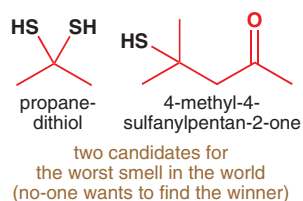
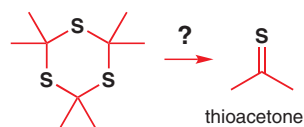
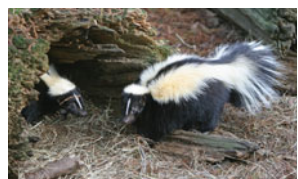
Among the 16 million that have been made, there are all kinds of molecules with amazingly varied properties. What do they look like? They may be crystalline solids, oils, waxes, plastics, elastics, mobile or volatile liquids, or gases. Familiar ones include sugar, a cheap natural compound isolated from plants as hard white crystals when pure, and petrol, a mixture of colourless, volatile, flammable hydrocarbons. Isooctane is a typical example and gives its name to the octane rating of petrol.



The compounds need not lack colour. Indeed we can soon dream up a rainbow of organic compounds covering the whole spectrum, not to mention black and brown. In this table we have avoided dyestuffs and have chosen compounds as varied in structure as possible.

Colour	Description	Compound	Structure
	red	dark red hexagonal plates	3-methoxybenzocycloheptatriene-2-one
	orange	amber needles	dichlorodicyanoquinone (DDQ)
	yellow	toxic yellow explosive gas	diazomethane
	green	green prisms with a steel-blue lustre	9-nitrosojulolidine
	blue	deep blue liquid with a peppery smell	azulene
	purple	deep blue gas condensing to a purple solid	nitrosotrifluoromethane

skunk spray contains:



Colour is not the only characteristic by which we recognize compounds. All too often it is their odour that lets us know they are around. There are some quite foul organic compounds too; the infamous stench of the skunk is a mixture of two thiols—sulfur compounds containing SH groups.

But perhaps the worst smell ever recorded was that which caused the evacuation of the German city of Freiburg in 1889. Attempts to make thioacetone by the cracking of trithioacetone gave rise to 'an offensive smell which spread rapidly over a great area of the town causing fainting, vomiting, and a panic evacuation...the laboratory work was abandoned'.

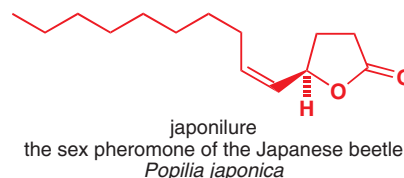
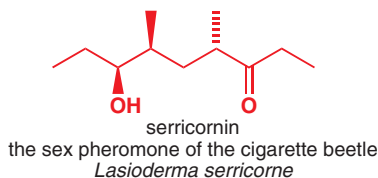
It was perhaps foolhardy for workers at an Esso research station to repeat the experiment of cracking trithioacetone south of Oxford in 1967. Let them take up the story. 'Recently we found ourselves with an odour problem beyond our worst expectations. During early experiments, a stopper jumped from a bottle of residues, and, although replaced at once, resulted in an immediate complaint of nausea and sickness from colleagues working in a building two hundred yards away. Two of our chemists who had done no more than investigate the cracking of minute amounts of trithioacetone found themselves the object of hostile stares in a restaurant and suffered the humiliation of having a waitress spray the area around them with a deodorant. The odours defied the expected effects of dilution since workers in the laboratory did not find the odours intolerable ... and genuinely denied responsibility since they were working in closed systems. To convince them otherwise, they were dispersed with other observers around the laboratory, at distances up to a quarter of a mile, and one drop of either acetone *gem*-dithiol or the mother liquors from crude trithioacetone crystallizations were placed on a watch glass in a fume cupboard. The odour was detected downwind in seconds.'

There are two candidates for this dreadful smell—propane dithiol (called acetone *gem*-dithiol above) or 4-methyl-4-sulfanylpentan-2-one. It is unlikely that anyone else will be brave enough to resolve the controversy.

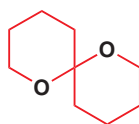
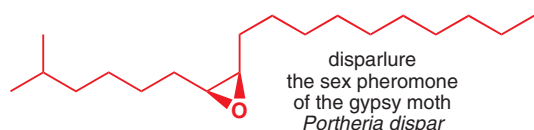
But nasty smells have their uses. The natural gas piped into homes contains small amounts of deliberately added sulfur compounds such as *tert*-butyl thiol ($(\text{CH}_3)_3\text{CSH}$). When we say small, we mean *very small*—humans can detect one part in 50,000,000,000 parts of natural gas.

Other compounds have delightful odours. To redeem the honour of sulfur compounds we must cite the truffle, which pigs can smell through a metre of soil and whose taste and smell is so delightful that truffles cost more than their weight in gold. Damascenones are responsible for the smell of roses. If you smell one drop you will be disappointed, as it smells rather like turpentine or camphor, but next morning you, and the clothes you were wearing, will smell powerfully of roses. Many smells develop on dilution.

Humans are not the only creatures with a sense of smell. We can find mates using all our senses, but insects cannot do this. They are small in a crowded world and they find those of the opposite sex of their own species by smell. Most insects produce volatile compounds that can be picked up by a potential mate in incredibly weak concentrations. Only 1.5 mg of serricornin, the sex pheromone of the cigarette beetle, could be isolated from 65,000 female beetles—so there isn't much in each beetle. Nevertheless, the slightest whiff of it causes the males to gather and attempt frenzied copulation. The sex pheromone of the beetle *Popilia japonica*, also given off by the females, has been made by chemists. As little as 5 μg (micrograms, note!) was more effective than four virgin females in attracting the males.



The pheromone of the gypsy moth, disparlure, was identified from a few μg isolated from the moths: as little as 2×10^{-12} g is active as a lure for the males in field tests. The three pheromones we have mentioned are available commercially for the specific trapping of these destructive insect pests.

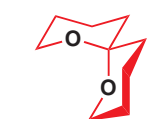


oleian
sex pheromone
of the olive fly
Bactrocera oleae



Gypsy moth

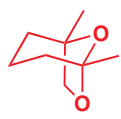
Don't suppose that the females always do all the work; both male and female olive flies produce pheromones that attract the other sex. The remarkable thing is that one mirror image of the molecule attracts the males while the other attracts the females! Mirror image isomers of a molecule called frontaline are also emitted by male elephants; female elephants can tell the age and appeal of a potential mate from the amount of each isomer he produces.



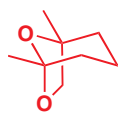
this mirror image
isomer attracts
male olive flies



this mirror image
isomer attracts
female olive flies



this mirror image
isomer smells of
young male elephant*

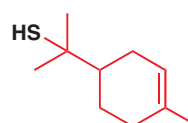


this mirror image
isomer smells of
old male elephant*

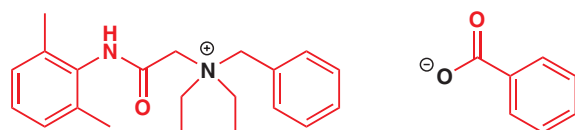
*if you are a
female elephant

What about taste? Take the grapefruit. The main flavour comes from another sulfur compound and human beings can detect 2×10^{-5} parts per billion of this compound. This is an almost unimaginably small amount equal to 10^{-4} mg per tonne or a drop, not in a bucket, but in a fairly large lake. Why evolution should have left us so extraordinarily sensitive to grapefruit, we leave you to imagine.

For a nasty taste, we should mention 'bittering agents', put into dangerous household substances like toilet cleaner to stop children drinking them by accident. Notice that this complex organic compound is actually a salt—it has positively charged nitrogen and negatively charged oxygen atoms—and this makes it soluble in water.



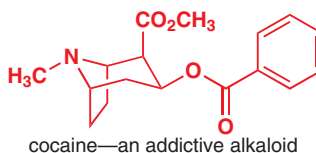
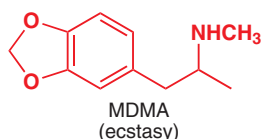
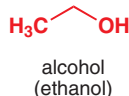
flavouring principle of grapefruit



'denatonium benzoate', marked as Bitrex
benzyl-diethyl[(2,6-xylyl-carbamoyl)methyl]ammonium benzoate

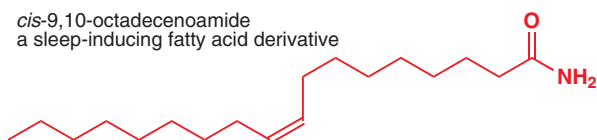


Other organic compounds have strange effects on humans. Various 'drugs' such as alcohol and cocaine are taken in various ways to make people temporarily happy. They have their dangers. Too much alcohol leads to a lot of misery and any cocaine at all may make you a slave for life.

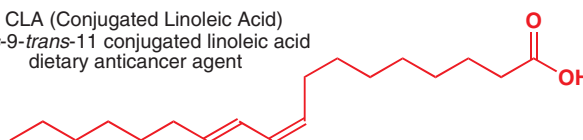


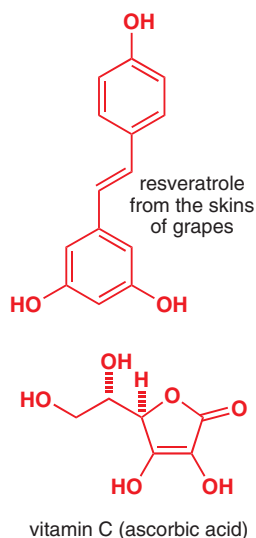
Again, let's not forget other creatures. Cats seem to be able to go to sleep anywhere, at any time. This surprisingly simple compound, isolated from the cerebrospinal fluid of cats, appears to be part of their sleep-control mechanism. It makes them, or rats, or humans fall asleep immediately.

cis-9,10-octadecenoamide
a sleep-inducing fatty acid derivative



CLA (Conjugated Linoleic Acid)
cis-9-*trans*-11 conjugated linoleic acid
dietary anticancer agent





This compound and disparlure (above) are both derivatives of fatty acids. Fatty acids in the diet are a popular preoccupation, and the good and bad qualities of saturates, monounsaturates, and polyunsaturates are continually in the news: one of the many dietary molecules reckoned to have demonstrable anticancer activity is CLA (conjugated linoleic acid), which is found in dairy products and also, most abundantly, you may be interested to know, in kangaroo meat.

Resveratrol is another dietary component with beneficial effects: it may be responsible for the apparent ability of red wine to prevent heart disease. It is a quite different sort of organic compound, with two benzene rings.

For a third edible molecule, how about vitamin C? This is an essential factor in your diet—that is why it is called a vitamin—and in the diet of other primates, guinea-pigs, and fruit bats (other mammals possess the biochemical machinery to make it themselves). The disease scurvy, a degeneration of soft tissues from which sailors on the long voyages of past centuries suffered, results from a lack of vitamin C. It also is a universal antioxidant, scavenging for rogue reactive radicals and protecting damage to DNA. Some people think an extra large intake may even protect against the common cold.

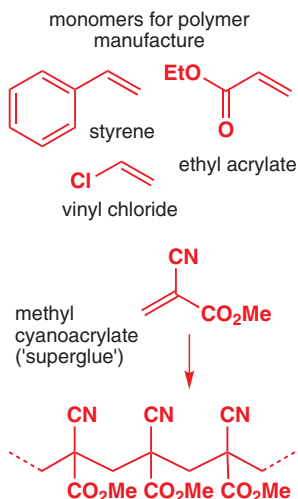
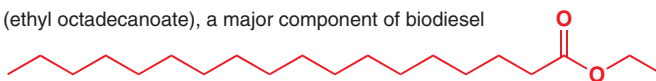
Organic chemistry and industry

Vitamin C is manufactured on a huge scale by Roche, a Swiss company. All over the world there are chemistry-based companies making organic molecules on scales varying from a few kilograms to thousands of tonnes per year. This is good news for students of organic chemistry: knowing how molecules behave and how to make them is a skill in demand, and it is an international job market.

The petrochemicals industry consumes huge amounts of crude oil: the largest refinery in the world, in Jamnagar, India, processes 200 million litres of crude oil every day. An alarmingly large proportion of this is still just burnt as fuel, but some of it is purified or converted into organic compounds for use in the rest of the chemical industry.

Some simple compounds are made both from oil and from plants. The ethanol used as a starting material to make other compounds in industry is largely made by the catalytic hydration of ethylene from oil. But ethanol is also used as a fuel, particularly in Brazil, where it is made by fermentation of sugar cane. Plants are extremely powerful organic chemical factories (with sugar cane being among the most efficient of all of them). Photosynthesis extracts carbon dioxide directly from the air and uses solar energy to reduce it to form less oxygen-rich organic compounds from which energy can be re-extracted by combustion. Biodiesel is made in a similar way from the fatty acid components of plant oils.

ethyl stearate (ethyl octadecanoate), a major component of biodiesel



Plastics and polymers take much of the production of the petrochemical industry in the form of monomers such as styrene, acrylates, and vinyl chloride. The products of this enormous industry are everything made of plastic, including solid plastics for household goods and furniture, fibres for clothes (over 25 million tonnes per annum), elastic polymers for car tyres, light bubble-filled polymers for packing, and so on. Worldwide 100 million tonnes of polymers are made per year and PVC manufacture alone employs over 50,000 people to make over 20 million tonnes per year.

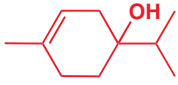
Many adhesives work by polymerization of monomers, which can be applied as a simple solution. You can glue almost anything with 'superglue', a polymer of methyl cyanoacrylate.

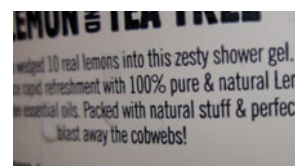
Washing-up bowls are made of the polymer polyethylene but the detergent you put in them belongs to another branch of the chemical industry—companies like Unilever and Procter and Gamble produce detergent, cleaners, bleaches, and polishes, along with soaps, gels, cosmetics, and shaving foams. These products may smell of lemon, lavender, or sandalwood but they too mostly come from the oil industry.

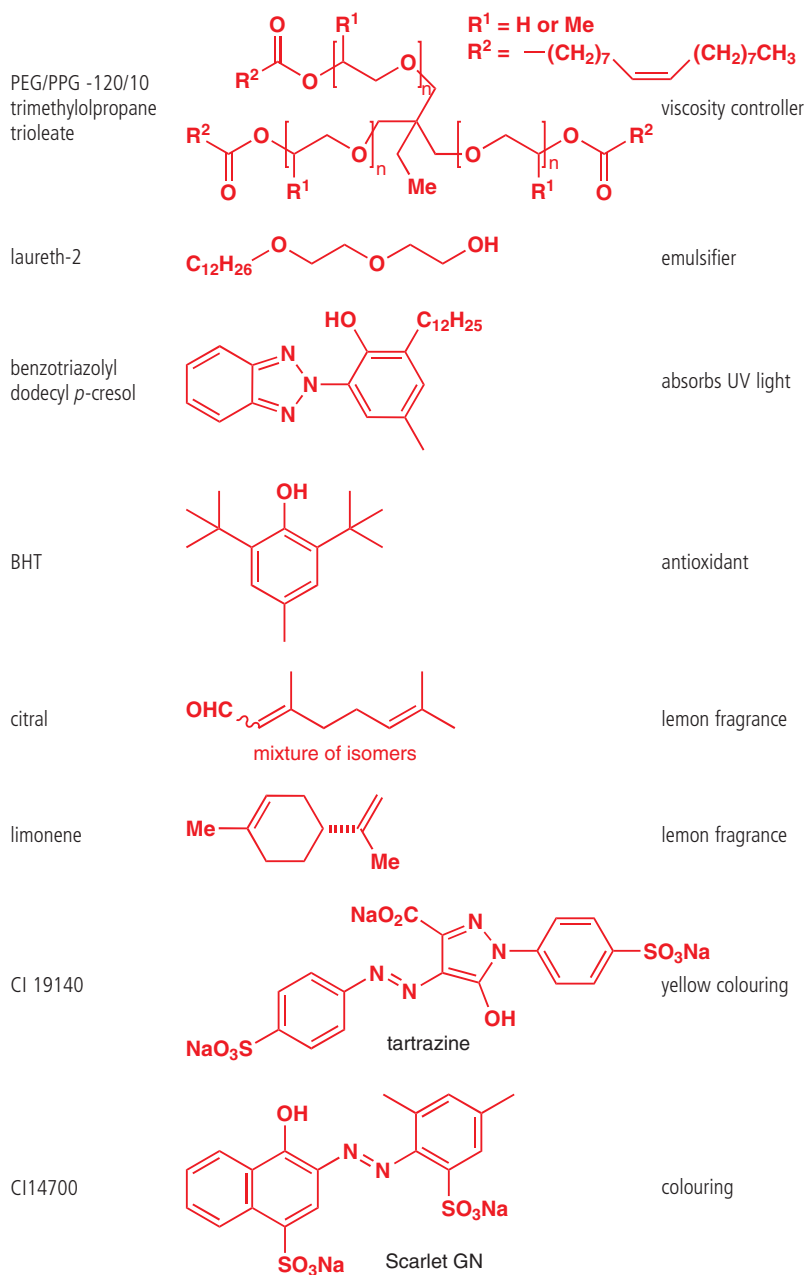
Products of this kind tend to underplay their petrochemical origins and claim affinity with the perceived freshness and cleanliness of the natural world. They also try to tell us, after a

fashion, what they contain. Try this example—the list of contents from a well-known brand of shower gel, which we are reassuringly told is ‘packed with natural stuff’ (including 10 ‘real’ lemons) and contains ‘100% pure and natural lemon and tea tree essential oils’.

It doesn’t all make sense to us, but here is a possible interpretation. We certainly hope this book will set you on the path of understanding the sense (and the nonsense!) of this sort of thing.

Ingredient	Chemical meaning	Purpose
aqua	$\text{H}-\text{O}-\text{H}$ water	solvent
sodium laureth sulfate	$\text{C}_{12}\text{H}_{25}(\text{OCH}_2\text{CH}_2)_n\text{OSO}_3\text{Na}$ typically $n = 3$	detergent
cocamide DEA	$\text{C}_{11}\text{H}_{23}\text{CON}(\text{CH}_2\text{CH}_2\text{OH})_2$	foaming agent
<i>Citrus medica limonum</i> peel oil	mainly  α -pinene	scent, appeal to customer
<i>Melaleuca alternifolia</i> leaf oil	mainly  terpinen-4-ol	scent, appeal to customer, possibly antiseptic
glycerin	$\text{HOCH}_2\text{CH}(\text{OH})\text{CH}_2\text{OH}$ glycerol	cosolvent; moisturizer; ensures smoothness
cocamidopropyl betaine	$\text{C}_{11}\text{H}_{24}\text{CON}(\text{CH}_2)_3\text{N}^+(\text{CH}_3)_2\text{CH}_2\text{COO}^-$	detergent and anti-electrostatic
sodium chloride	NaCl	control solubility of Na^+ -based detergents
lactic acid	$\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$	acidifier
styrene acrylates copolymer	$\left[\text{Ph}-\text{CH}_2-\text{CH}(\text{CO}_2\text{R}) \right]_n \left[\text{R}-\text{CH}_2-\text{CH}(\text{CO}_2\text{R}) \right]_m$ $\text{R} = \text{Me or H}$	film former
tetrasodium glutamate diacetate	$\text{NaO}_2\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}(\text{CO}_2\text{Na})-\text{CH}_2-\text{N}(\text{CH}_2\text{CO}_2\text{Na})_2$	chelator, to prevent formation of insoluble scum in hard water
sodium benzoate	$\text{C}_6\text{H}_5\text{CO}_2\text{Na}$	preservative

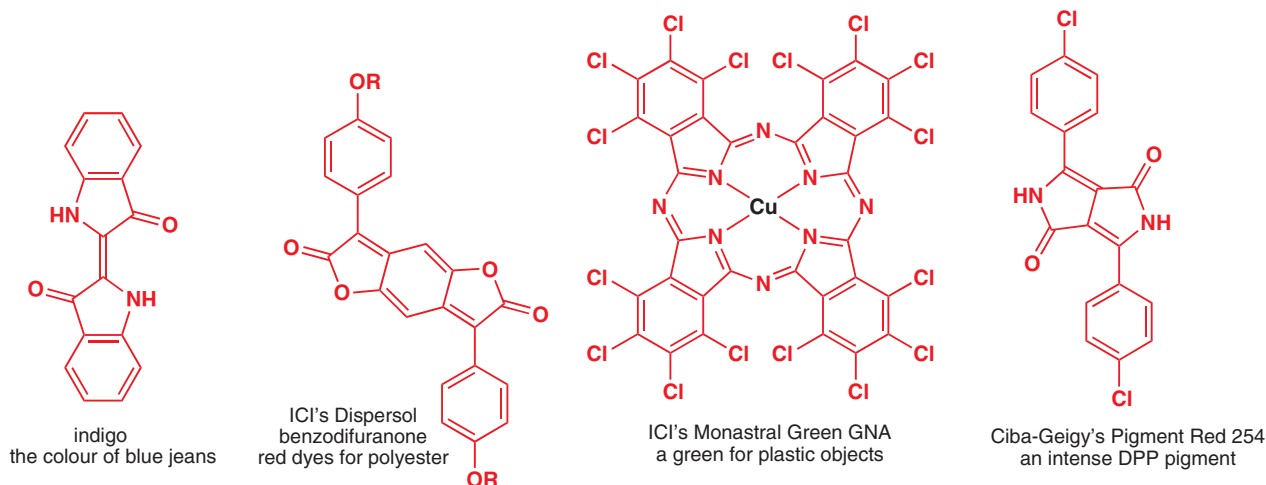




The particular detergents, surfactants, acids, viscosity controllers, and so on are chosen to blend together to give a smooth gel. The result should feel, smell, and look attractive and work as an effective detergent and shampoo (some of the compounds are added for their moisturizing and anti-electrostatic effect on hair). The yellow colour and lemon scent are considered fresh and clean by the customer. Several of the ingredients are added as pure compounds; the ones which aren't are mixtures of isomers or polymers; the most impure are the mixtures of hydrocarbons referred to as the 'pure and natural' essential oils. Is it 'packed with natural stuff'? Indeed it is. It all comes from natural sources, the principal one being decomposed carboniferous forests trapped for millions of years underground.

The coloration of manufactured goods is a huge business, with a range of intense colours required for dyeing cloth, colouring plastic and paper, painting walls, and so on. Leaders in this area are companies such as Akzo Nobel, which had sales of €14.6 bn in 2010. One of the most commonly used dyestuffs is indigo, an ancient dye that used to be isolated from plants but is now made from petrochemical feedstocks. It is the colour of blue jeans. More modern

dyestuffs can be represented by the benzodifuranones developed by ICI, which are used for colouring synthetic fabrics like polyesters (red), the phthalocyanine–metal complexes (typically blue or green), or the ‘high-performance’ red pigment DPP (1,4-diketopyrrolo[3,4-*c*]pyrroles) series developed by Ciba-Geigy.

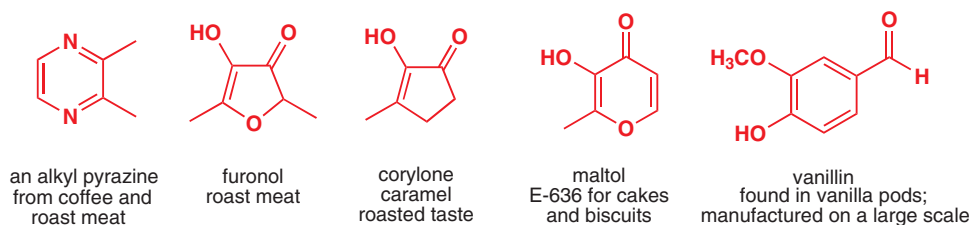


The scent of the shower gel above came from a mixture of plant extracts with the pure compound (in fact a mixture of two isomers) citral. The big fragrance and flavouring companies (such as Firmenich, International Flavors and Fragrances, and Givaudan) deal in both naturals and synthetics—‘naturals’ are mixtures of compounds extracted from plants—leaves, seeds, and flowers. ‘Synthetics’ are single compounds, sometimes present in plant-derived sources and sometime newly designed molecules, which are mixed with each other and with ‘naturals’ to build up a scent. A typical perfume will contain 5–10% fragrance molecules in an ethanol/water (about 90:10) mixture. So the perfumery industry needs a very large amount of ethanol and, you might think, not much perfumery material. In fact, important fragrances like jasmine are produced on a >10,000 tonnes per annum scale. The cost of a pure perfume ingredient like *cis*-jasmone (p. 2), the main ingredient of jasmine, may be several hundred pounds, dollars, or euros per gram.

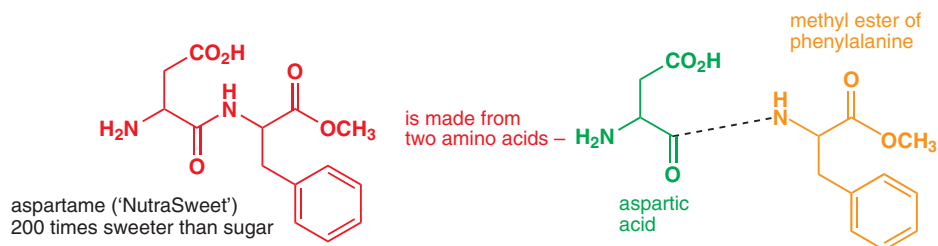
The world of perfumery

Perfume chemists use extraordinary language to describe their achievements: ‘PacoRabanne pour homme was created to reproduce the effect of a summer walk in the open air among the hills of Provence: the smell of herbs, rosemary and thyme, and sparkling freshness with cool sea breezes mingling with warm soft Alpine air. To achieve the required effect, the perfumer blended herbaceous oils with woody accords and the synthetic aroma chemical dimethylheptanol, which has a penetrating but indefinable freshness associated with open air or freshly washed linen.’

Chemists produce synthetic flavourings such as ‘smoky bacon’ and even ‘chocolate’. Meaty flavours come from simple heterocycles such as alkyl pyrazines (present in coffee as well as roast meat) and furonol, originally found in pineapples. Compounds such as corylone and maltol give caramel and meaty flavours. Mixtures of these and other synthetic compounds can be ‘tuned’ to taste like many roasted foods from fresh bread to coffee and barbecued meat. Some flavouring compounds are also perfumes and may also be used as an intermediate in making other compounds. Vanillin is the main component of the flavour of vanilla, but is manufactured on a large scale for many other uses too.

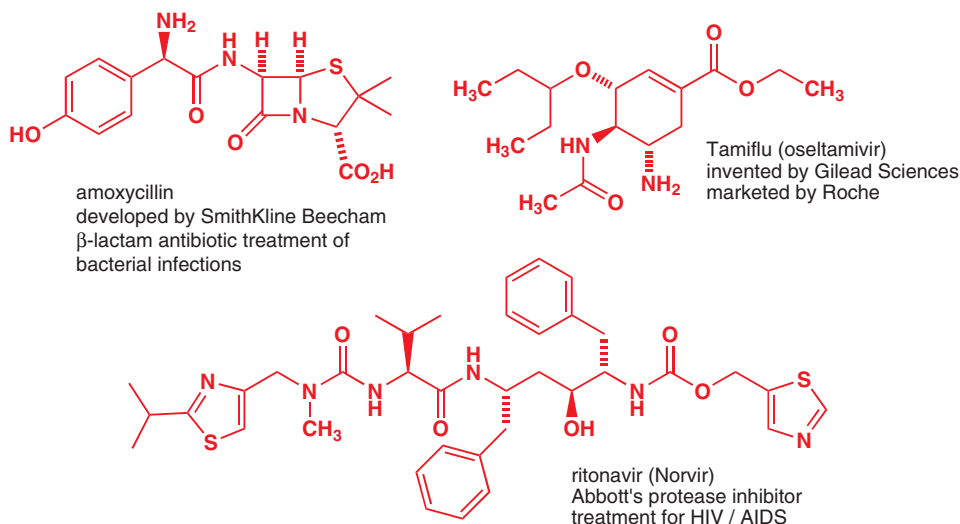


Food chemistry includes much larger-scale items than flavours. Sweeteners such as sugar itself are isolated from plants on an enormous scale. You saw sucrose on p. 3, but other sweeteners such as saccharin (discovered in 1879!) and aspartame (1965) are made on a sizeable scale. Aspartame is a compound of two of the natural amino acids present in all living things and over 10,000 tonnes per annum are made by the NutraSweet company.

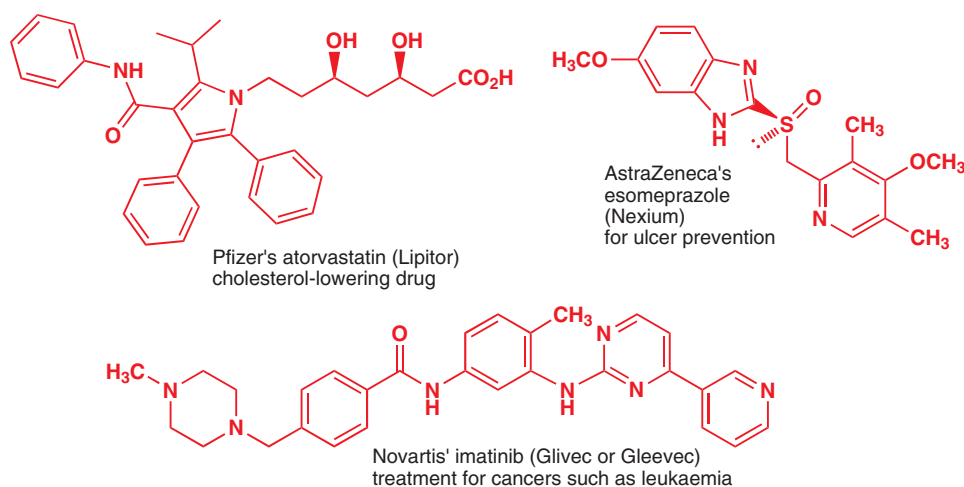


➡ The story of Tamiflu and how the ingenuity of chemists ensures a constant supply is related at the other end of this book, in Chapter 43.

One of the great revolutions of modern life has been the expectation that humans will survive diseases because of a specifically designed treatment. In the developed world, people live to old age because infections which used to kill can now be cured or kept at bay. Antibiotics are our defence against bacteria, preventing them from multiplying. One of the most successful of these is Beecham amoxycillin, which was developed by SmithKline. The four-membered ring at the heart of the molecule is the β -lactam, which targets the disease-causing bacteria. Medicinal chemists also protect us from the insidious threat of viruses which use the body's own biochemistry to replicate. Tamiflu is a line of defence against the ever-present danger of a flu epidemic, while ritonavir is one of the most advanced drugs designed to prevent replication of HIV and to slow down or prevent the onset of AIDS.

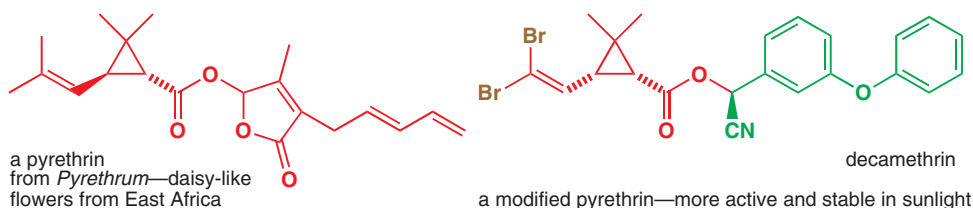


The best-selling current drugs are largely designed to address the human body's own failings. Sales of Lipitor and Nexium both topped \$5bn in 2009, figures which serve to illustrate the financial scale of developing safe and effective new treatments. Lipitor is one of the class of drugs known as statins, widely prescribed to control cholesterol levels in older people. Nexium is a proton pump inhibitor, which works to reduce peptic and duodenal ulcers. Sales of Glivec (developed by Novartis and introduced in 2001) are far smaller, but to those suffering from certain cancers such as leukaemia it can be a lifesaver.

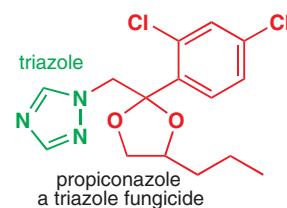


We cannot maintain our present high density of population in the developed world, nor deal with malnutrition in the developing world unless we preserve our food supply from attacks by insects and fungi and from competition by weeds. The world market for agrochemicals produced by multinationals such as Bayer CropScience and Syngenta is over £10bn per annum divided between herbicides, fungicides, and insecticides.

Many of the early agrochemicals were phased out as they were persistent environmental pollutants. Modern agrochemicals have to pass stringent environmental safety tests. The most famous modern insecticides are modelled on the plant-derived pyrethrins, stabilized against degradation in sunlight by chemical modification (the brown and green portions of decamethrin) and targeted to specific insects on specific crops. Decamethrin has a safety factor of >10,000 for mustard beetles over mammals, can be applied at only 10 grams per hectare (about one level tablespoon per football pitch), and leaves no significant environmental residue.



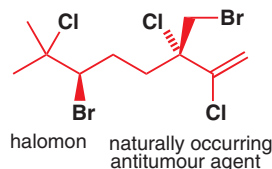
As you learn more chemistry, you will appreciate how remarkable it is that Nature should produce the three-membered rings in these compounds and that chemists should use them in bulk compounds to be sprayed on crops in fields. Even more remarkable in some ways are the fungicides based on a five-membered ring containing three nitrogen atoms—the triazole ring. These compounds inhibit an enzyme present in fungi but not in plants or animals. Fungal diseases are a real threat: as in the Irish potato famine of the nineteenth century, the various fungal blights, blotches, rots, rusts, smuts, and mildews can overwhelm any crop in a short time.



Organic chemistry and the periodic table

All the compounds we have shown you are built up on hydrocarbon (carbon and hydrogen) skeletons. Most have oxygen and/or nitrogen as well; some have sulfur and some phosphorus, and maybe the halogens (F, Cl, Br, and I). These are the main elements of organic chemistry.

fialuridine
antiviral compound

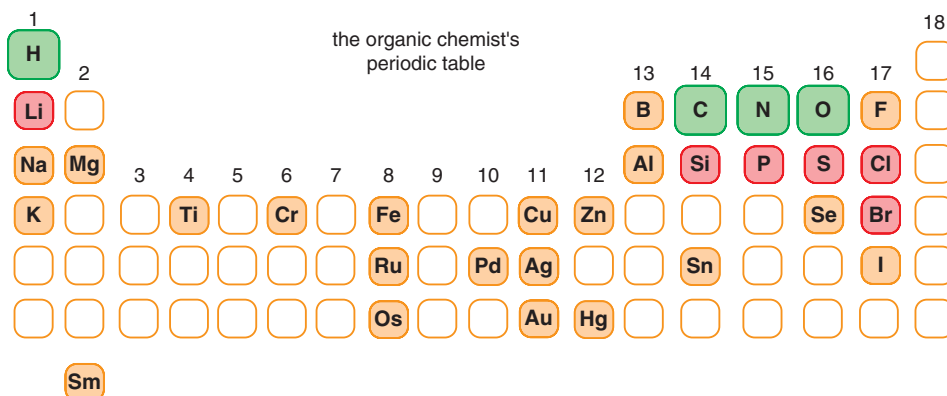


➡ We will devote whole chapters to the organic chemistry of S, P, and Si (Chapter 27) and to the transition metals, especially Pd (Chapter 40).



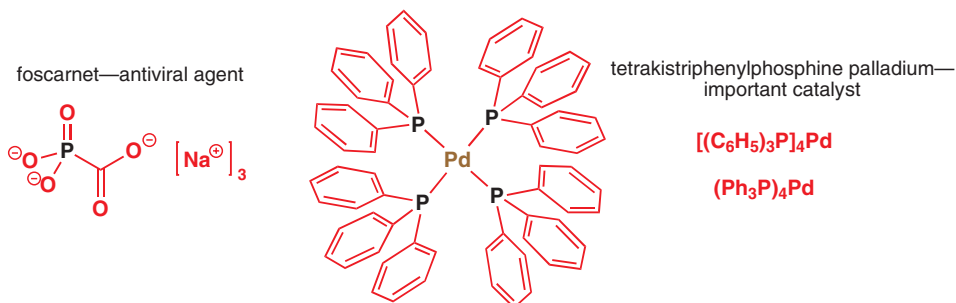
The halogens also appear in many life-saving drugs. Antiviral compounds such as fialuridine (which contains both F and I, as well as N and O) are essential for the fight against HIV and AIDS. They are modelled on natural compounds from nucleic acids. The naturally occurring cytotoxic (antitumour) agent halomon, extracted from red algae, contains Br and Cl.

The organic chemist's periodic table would have to emphasize all of these elements and more—the table below highlights most of those elements in common use in organic reactions. New connections are being added all the time—before the end of the last century the organic chemistry of ruthenium, gold, and samarium was negligible; now reagents and catalysts incorporating these metals drive a wide range of important reactions.



■ You will certainly know something about the periodic table from your previous studies of chemistry. A full Periodic Table appears on pp. 1184–1185 of this book, but basic knowledge of the groups, which elements are metals, and where the elements shown in this table appear will be helpful to you.

So where does inorganic chemistry end and organic chemistry begin? Would you say that the antiviral compound foscarnet was organic? It is a compound of carbon with the formula CPO_5Na_3 but it has no C–H bonds. And what about the important reagent tetrakis (triphenylphosphine)palladium? It has lots of hydrocarbon—12 benzene rings in fact—but the benzene rings are all joined to phosphorus atoms that are arranged in a square around the central palladium atom, so the molecule is held together by C–P and P–Pd bonds, not by a hydrocarbon skeleton. Although it has the very organic-looking formula $\text{C}_{72}\text{H}_{60}\text{P}_4\text{Pd}$, many people would say it is inorganic. But is it?



The answer is that we don't know and we don't care. Strict boundaries between traditional disciplines are undesirable and meaningless. Chemistry continues across the old boundaries between organic chemistry and inorganic chemistry, organic chemistry and physical chemistry or materials, or organic chemistry and biochemistry. Be glad that the boundaries are indistinct as that means the chemistry is all the richer. This lovely molecule $(\text{Ph}_3\text{P})_4\text{Pd}$ belongs to *chemistry*.

Organic chemistry and this book

We have told you about organic chemistry's history, the types of compounds it concerns itself with, the things it makes, and the elements it uses. Organic chemistry today is the study of the structure and reactions of compounds in nature, of compounds in the fossil reserves such as coal and oil, and of those compounds that can be made from them. These compounds will usually be constructed with a hydrocarbon framework but will also often have atoms such as O, N, S, P, Si, B, halogens, and metals attached to them. Organic chemistry is used in the making of plastics, paints, dyestuffs, clothes, foodstuffs, human and veterinary medicines, agrochemicals, and many other things. Now we can summarize all of these in a different way.

● The main components of organic chemistry as a discipline are:

- **structure determination**—how to find out the structures of new compounds even if they are available only in invisibly small amounts
- **theoretical organic chemistry**—how to understand these structures in terms of atoms and the electrons that bind them together
- **reaction mechanisms**—how to find out how these molecules react with each other and how to predict their reactions
- **synthesis**—how to design new molecules, and then make them
- **biological chemistry**—how to find out what Nature does and how the structures of biologically active molecules are related to what they do.

This book is about all these things. It is about the structures of organic molecules and the reasons behind those structures. It is about the shapes of these molecules and how the shape relates to their function, especially in the context of biology. It explains how these structures and shapes are discovered. It tells you about the reactions the molecules undergo and, more importantly, how and why they behave in the way they do. It tells you about nature and about industry. It tells you how molecules are made and how you too can think about making molecules.

This is the landscape through which you are about to travel. And, as with any journey to somewhere new, exciting, and sometimes challenging, the first thing is to make sure you have at least some knowledge of the local language. Fortunately the language of organic chemistry couldn't be simpler: it's all pictures. The next chapter will get us communicating.

Further reading

One interesting and amusing book you might enjoy is B. Selinger, *Chemistry in the Marketplace*, 5th edn, Harcourt Brace, Sydney, 2001.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>